Cross-Modal Drone Detection: A Self-Improving, Deployable Detection Pipeline

Mini preview — structure & story for review (not the final draft)

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Abstract

This document is a *compressed preview* of the restructured thesis: the narrative spine, the IMRAD skeleton, and the empirical tables, with current-best numbers tagged [FINAL] (locked) or [PROVISIONAL] (to be replaced by the full-frame run). Read it to approve the *shape of the argument*; the full chapters are expanded only after sign-off.

Thesis in one paragraph. We build and validate a *deployable* dual-modality (RGB + thermal-IR) drone-detection *system* — a fusion pipeline with an operator GUI — and we show it is produced and *maintained* by a self-improving loop: a human-in-the-loop (HITL) data engine that co-develops the detectors, and a statistics-first feature-space diagnostic ("Model MRI") that derives cheap, robust components and catches regressions before they ship. The headline is the *system and the method that sustains it* (contributions C1–C3); cross-modal grayscale transfer, a 28-point scoring-rule audit, and large speed wins are supporting *findings*.

1 Introduction

Problem. Drone detection is reported on saturated academic benchmarks, but a detector that wins a benchmark is not a system that survives deployment. Real operation adds (i) confusers — birds, aircraft, helicopters — that a recall-tuned detector hallucinates as drones; (ii) hard surfaces (low-resolution CCTV, distant/small targets) where benchmark recall collapses; and (iii) degraded sensing (a thermal channel that is really a desaturated RGB feed). The gap between *benchmark score* and *deployable system* is the problem this thesis addresses.

Research questions.

RQ1 Can a learned, leakage-aware pipeline suppress out-of-distribution confuser false positives without sacrificing drone recall?

RQ2 What is the in-distribution cost of that pipeline, and how does it depend on the surface (clean benchmark vs. hard CCTV)?

RQ3 Does a thermal-trained detector transfer to a degraded *grayscale* channel, and can the same diagnostic make that transfer reliable?

Contributions.

C1 — The pipeline + GUI. A deployable dual-modality detection pipeline (per-modality detect → trust-routing classifier → feature-space verifier → temporal alert-gate) with an operator GUI. The *artifact*.

C2 — The Model MRI. A statistics-before-training, feature-space diagnostic (LDA/PCA/ANOVA/AURO that tells us *what will work before we train*. It auto-derives the cheap MLP verifier and the leakage-aware trust router, and it catches regressions.

C3 — The HITL data engine. A human-in-the-loop label-reviewer that co-develops the detectors and provides the discipline that catches a real regression (the IR V5 event).

Supporting *findings* (not the spine): cross-modal grayscale transfer (RQ3); a scoring-rule audit; and 37–404× component speed-ups.

2 Background & Related Work

[expand later]: benchmark saturation (Anti-UAV); confuser robustness; cross-modal transfer; HITL annotation; feature-space diagnostics. One paragraph each, positioning C1–C3 against prior work.

3 Methodology

3.1 The system (C1)

The pipeline scores each modality independently and never unions their evidence. Stages: (1) **detectors** — RGB `ft4` and thermal-IR `v3b`; (2) a **trust-routing classifier** (`robust8`) that decides, per frame, which modality to believe (trust-RGB / trust-IR / trust-both / reject); (3) a per-detection **verifier** (`mlp_v5` for RGB, `mlp_v5_ir_aligned` for thermal *and* grayscale) that drops confuser detections using the detector's own ROI features; (4) a **temporal alert-gate** (segment-level 2-of-3 voting, mirrored from the GUI). **GUI**: [real screenshot needed] — operator view, per-modality tracks, alert gate, grayscale fallback path.

3.2 The HITL data engine (C3)

Detectors are co-developed with a label-reviewer in the loop: every mined batch is human-vetted before it enters training. Section 4.1 shows what happens when that discipline is *bypassed* (the IR V5 regression).

3.3 The Model MRI (C2)

Before any training run we diagnose separability in feature space. The detector's own ROI features *already* separate drones from confusers — LDA accuracy 0.952 (RGB) / 0.981 (IR) [FINAL] and a dominant ANOVA effect ($F = 42{,}346$) [FINAL] — which is the hypothesis that justifies a *cheap MLP* verifier instead of re-processing every crop with a CNN. The same diagnostic shows the grayscale channel is *alignable* to the thermal feature space: per-modality z-alignment lifts held-out gray→thermal discrimination from AUROC 0.500 to 0.919 [FINAL]. **This is the through-line:** a hypothesis in feature space → figures → a derived component → a pipeline win.

3.4 Component design as a three-version argument

Each component is presented as three versions — a *sad* path, a *happy* path, and the step that *justifies the next thing*:

- **RGB detector**: baseline / retrained_v2 (OOD recall collapse) / `ft4`. The collapse cannot be fixed at the detector → motivates the verifier.
- **IR detector**: V2 / V5 (HITL regression) / `v3b`. Motivates HITL discipline (C3).

- **Trust router**: `sa32` + `fusion_no_fn` (32–40 feat, expensive, over-rejects) / `robust8` (8 feat). Motivates leakage-aware feature selection (C2).
- **Verifier**: MobileNetV3 patch CNN (slow, OOD-weak) / `mlp_v5` (distilled). Motivates the MRI: are the features separable? Yes (LDA ≈ 0.95) $\rightarrow$ distilled MLP wins.

3.5 Evaluation design

Surfaces (7 + grayscale): RGB test, IR test, Anti-UAV (paired), Svanström (paired), selcom CCTV, RGB confusers, IR confusers, and a grayscale path. Birds are a *category* inside the confuser sets, not a dataset. **Image size:** 640 default; 1280 only for low-native-resolution sources (Svanström 640×512, selcom CCTV). **Scoring is per-modality, never unioned:** trust-RGB scores RGB detections vs. RGB GT only (IR ground truth is *excluded*, not counted as a miss); only reject-both counts both as misses. Svanström uses IoP@0.5, Anti-UAV IoU@0.5; confusers have no GT (false positives / fire-rate only). **Two tiers:** Tier-1 is a ∼4k *strided* screen (per-frame numbers, this preview); Tier-2 is the full consecutive-frame run (final numbers + temporal). Temporal voting needs consecutive frames, so it is a placeholder until Tier-2.

4 Empirical Evaluation

Metric status. Part A (detectors), the IR arc, the MRI numbers, and speed are [FINAL] (at-standard, recorded). Part B/C/D pipeline cells are [PROVISIONAL] and will be replaced by the 4k strided screen and then the full-frame run; the structure does not change, only the digits.

4.1 Part A — Per-model, in-domain (bare detectors)

Table 1: RGB detector, three versions. In-domain all three are competitive; the differences are *out-of-distribution*. `ft4`'s low *bare* Svanström F1 is precision traded for recall (R= 0.914) — the pipeline recovers it (Part B). [FINAL] except where noted.

surface (imgsz, rule)	baseline	retrained_v2	`ft4` (prod)
rgb_dataset_test (640, IoU)	0.942	**0.949**	0.929
Svanström (1280, IoP)	**0.950**	0.462	0.596 (bare)
selcom CCTV (1280, IoP)	0.145	0.007	**0.591**
Anti-UAV (saturated)	0.994	0.994	0.986

Reading. There is *no single best bare detector*: baseline wins Svanström, `ft4` is the only one that holds CCTV (0.591 vs. 0.145), and retrained_v2 wins nothing out-of-domain (Svanström 0.462, selcom 0.007 — a single true positive in 295). Crucially, retrained_v2 is *fine in-domain* (0.949): a model chosen on the own-split benchmark is the one that fails in deployment. [FINAL] This is the entire motivation for the pipeline (C1): the best *deployable* configuration is not the best bare detector, it is `ft4` + pipeline.

4.2 Part B — Full-pipeline ablation (paired surfaces)

On the paired drone-positive surfaces (Anti-UAV, Svanström) we ablate bare $\rightarrow$ +router $\rightarrow$ +verifier $\rightarrow$ full cascade, scored per-modality. The question is whether precision rises *without* losing recall.

Recall is preserved (R 0.914→0.838) while precision roughly doubles (0.443→0.903). The trust router (`robust8`) adds modality selection on top; mini-run thermal F1 = 0.979 [PROVISIONAL]. [4k]: fill the full bare/+C/+F/full grid for Anti-UAV and Svanström from the unified cache.

Table 2: IR detector evolution (own test split, 640, IoU). V5 is a HITL regression: a large positive batch bypassed per-batch review, so unlabelled drones scored as false positives. The HITL discipline (C3) is what recovered it. [FINAL]

version	P	R	F1	
V2	0.661	0.406	0.503	early, weak recall
V5	–	–	0.737	HITL regression
`v3b` (prod)	0.957	0.977	**0.967**	corrective FT

Table 3: Svanström pipeline (IoP@0.5). The verifier recovers `ft4`'s bare precision deficit. [PROVISIONAL] (mini-run / at-standard head-to-head; refresh from 4k then full).

stage	P	R	F1
`ft4` bare	0.443	0.914	0.596
+ patch verifier (MobileNet)	0.686	0.872	0.768
+ `mlp_v5` verifier	**0.903**	0.838	**0.869**

4.3 Part C — Confuser false-positive reduction (OOD benefit)

On confuser surfaces (no drones present, so every surviving detection is a false positive) the verifier is the benefit. `mlp_v5` cuts RGB-confuser false positives 216 → 16 (∼13×) [PROVISIONAL] and, in the thermal/grayscale-aligned feature space, held-out CBAM F1 rises 0.699 → 0.841 with false positives 48 → 13 [FINAL]. The MobileNet patch CNN achieves comparable suppression but 37–72× slower (Section 4.5). [4k]: rgb / ir / gray confuser FP grid (bare → +mlp → +patch) from the cache.

4.4 Part D — Grayscale cross-modal transfer (finding, RQ3)

We report *only* the working configuration: the thermal detector `v3b` run on a *grayscale* RGB feed, plus the aligned verifier, with the (thermal-trained) router bypassed. On held-out grayscale confusers the aligned verifier cuts false positives 325 → 12 (96%) while keeping recall within ∼5 points of bare [FINAL], and the MRI alignment that makes this possible lifts gray→thermal AUROC 0.500 → 0.919 (Section 3). The degenerate configurations (router on grayscale; a grayscale-specific verifier that over-vetoes) are mentioned in one honest sentence and not tabulated.

4.5 Speed

Table 4: Component cost (the cheap-by-design payoff of C2). [FINAL] (bench).

component	sad	happy	speed-up
trust router	`fusion_no_fn` 38.3 ms/frame	`robust8` 0.095 ms/frame	∼404×
verifier	patch 59–112 ms/det	`mlp_v5` 1.3–2.1 ms/det	∼37–72×

Total pipeline overhead is ∼1–4%. [wire bench numbers to knowledge/ledger before final].

4.6 Threats to validity

- **Strided screen.** Tier-1 numbers are a strided sample; temporal continuity is not yet measured. Tier-2 (full consecutive frames) resolves both.

- **Saturated benchmark.** Anti-UAV is saturated; it is a sanity floor, not a discriminator.
- **Per-modality scoring choice.** Excluding the untrusted modality's GT is a deliberate rule; we report and justify it rather than silently unioning (the 28-point audit).
- **Train/eval provenance.** Components are scored by pipeline *effect*, not their own held-out accuracy, because those splits are not cross-comparable.

5 Conclusion

A benchmark detector is not a deployable system. We close that gap with a per-modality pipeline and operator GUI (C1), kept honest and cheap by a feature-space diagnostic (C2) and a human-in-the-loop data engine (C3). The components are not individually dominant — no single bare detector wins every surface — but their *composition* is the only configuration deployable across clean benchmarks, hard CCTV, confuser-rich scenes, and a degraded grayscale channel.

Review checklist for the author. (1) Is the A+D spine (system as headline, grayscale as finding) what you want? (2) Is the Part A→B framing of the "no single best detector" tension acceptable? (3) Sign-off gates the full expansion into the Overleaf thesis; Tier-2 then swaps every [PROVISIONAL] digit.